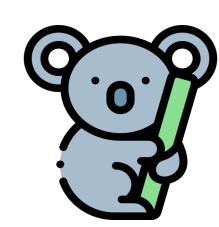


Land-Use Change Prediction in Madagascar: Public Health Implication

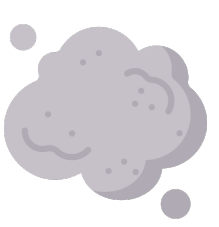
yuval bloch, charls nunn, kayla kaoufman, kacy ring, shai philosoph

importance

Land use represents the fundamental interface between humanity and nature. it the main cause of:



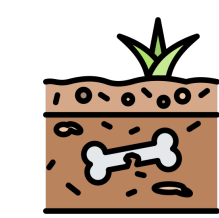
85%of endangered species cases



almost aquarter of greenhouse gases.



the emergence of 27% of vector-borne zoonotic diseases (VBZD)



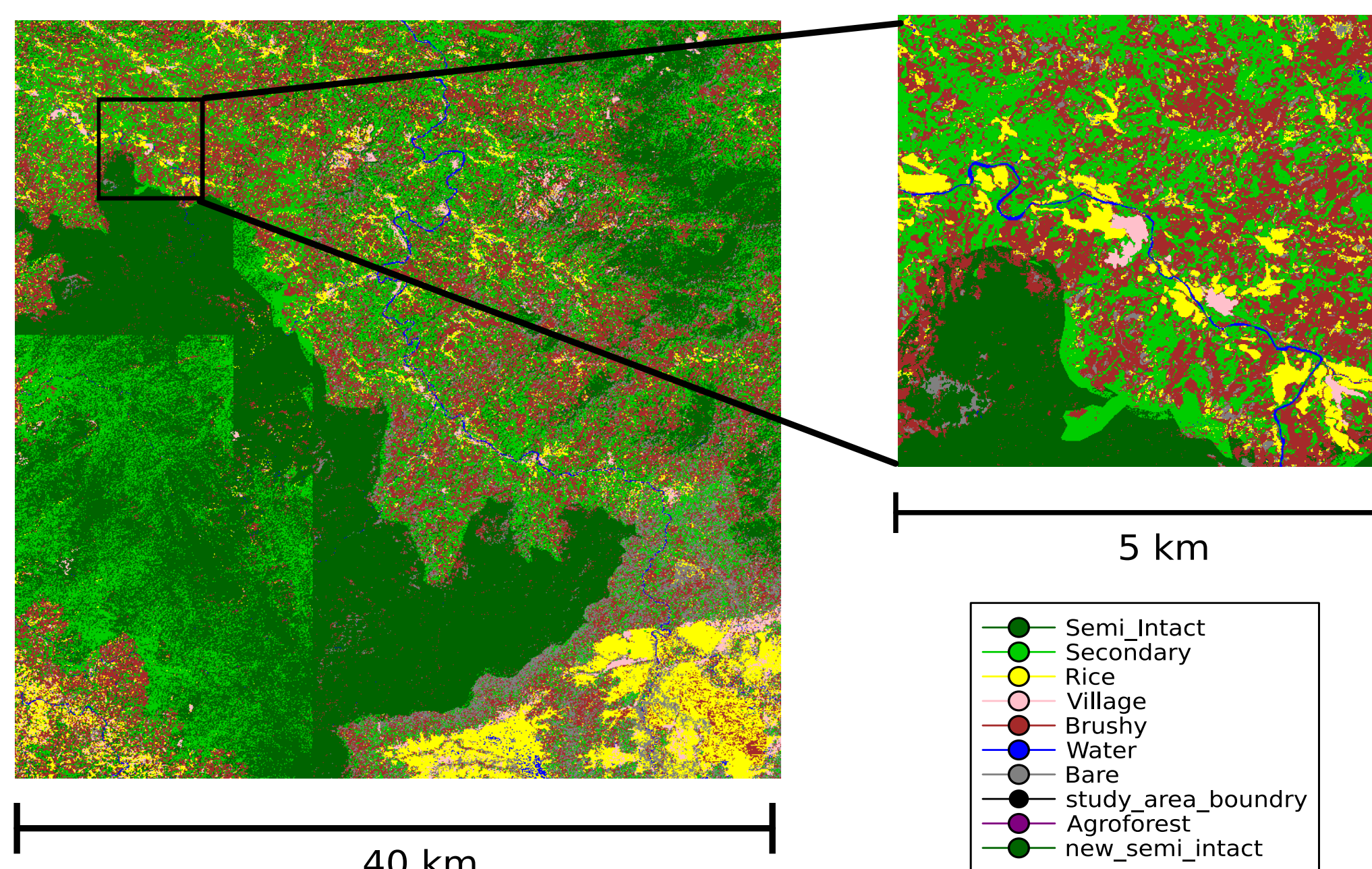
leading cause of soil loss

this interface is highlited in small holder farms in the troipcs where lack of infrastructure, make the population vulnerable to environmental change

chlanges

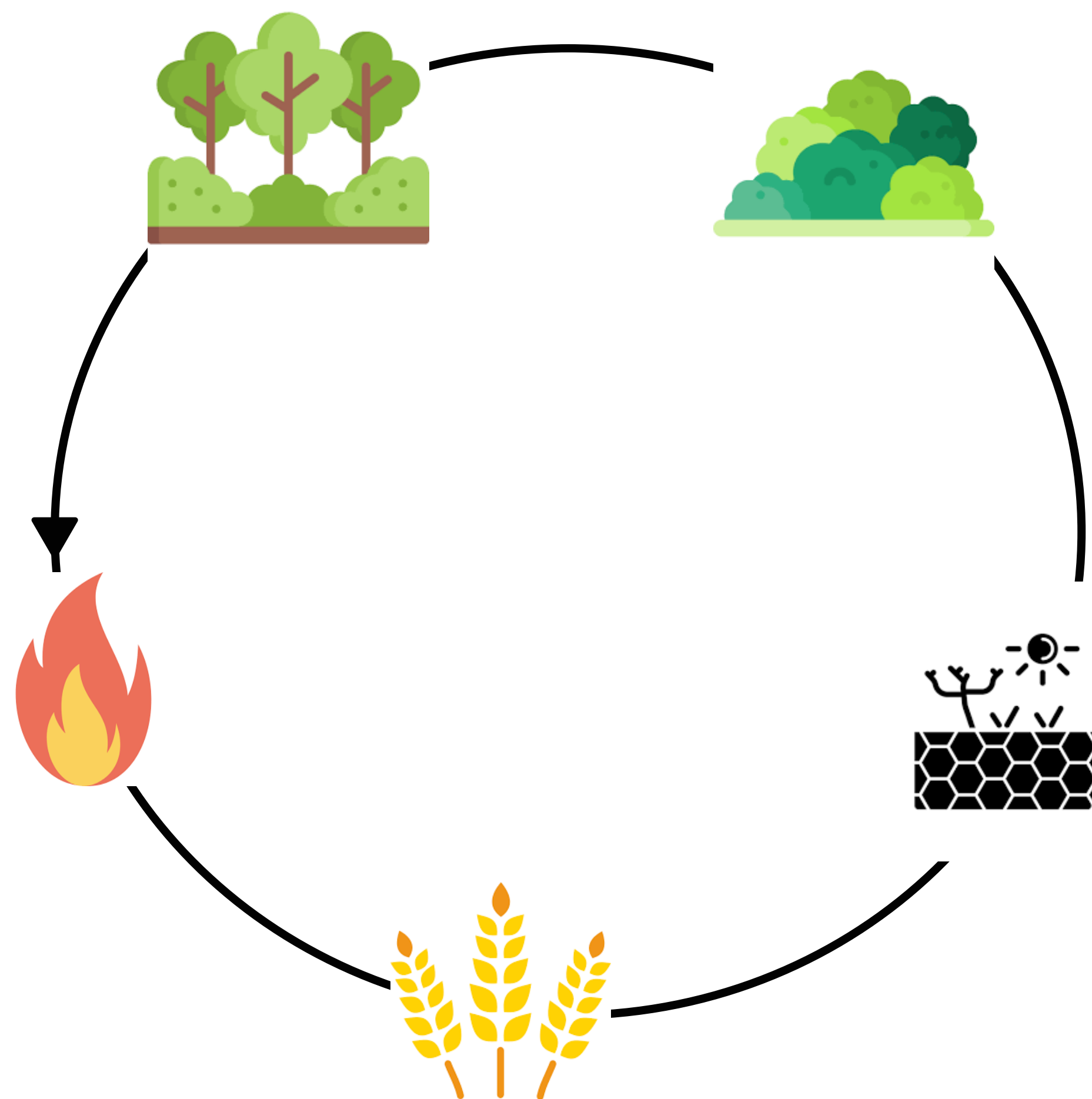
hertoginaty across spitial scale

in many underdeloped country where the land shred between many smallholder farms the land use map are complex and hetorgiuns even in small scales



hertoginaty across temporal scale

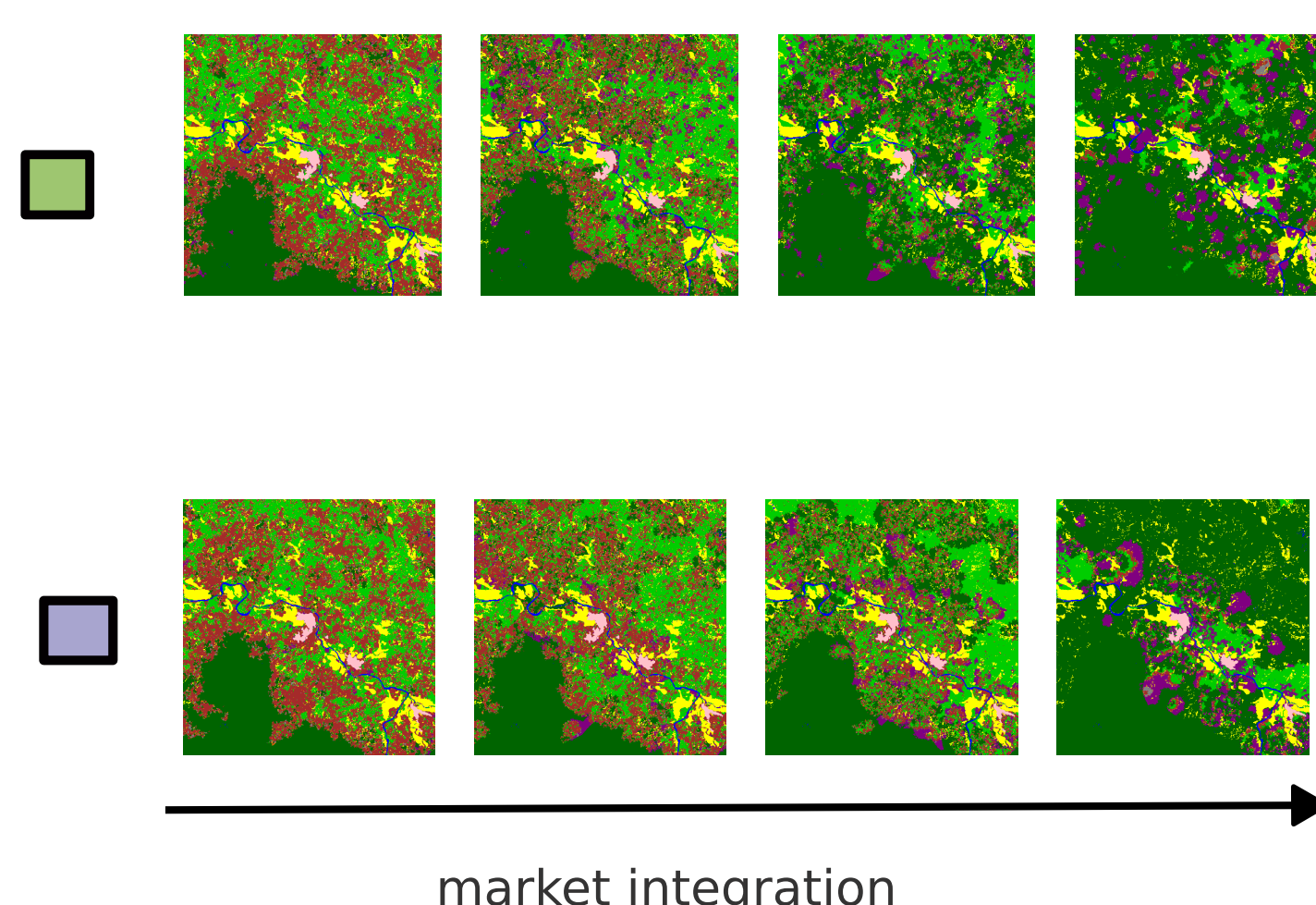
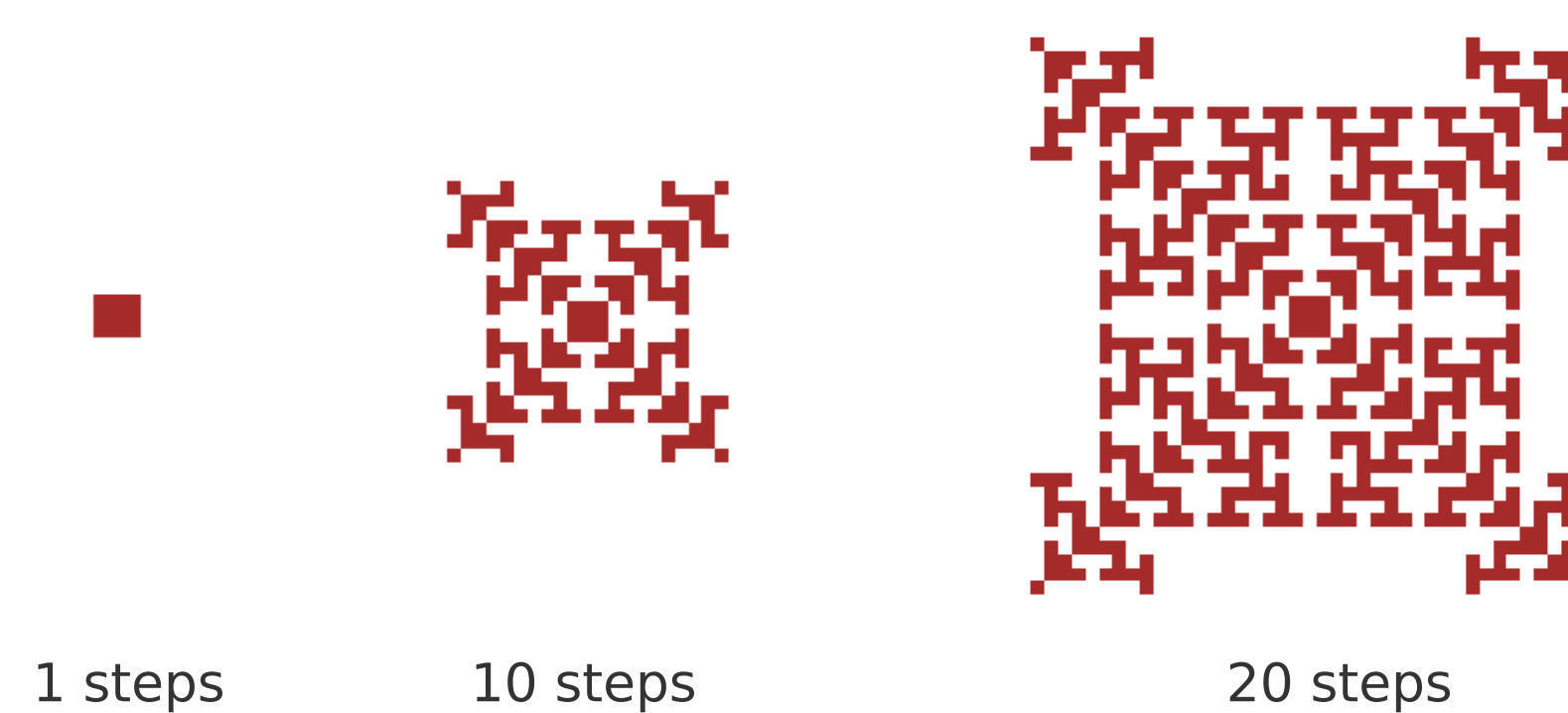
in shifting cultivation the same land plot pase trough several pahge of land cover



methods

celoluer autotmta (CA): simple rules for complex process

for a simple example let we have two cell type green and white, white cell became green if and only if it have excaly one green nghborrs including diagonal, as you can see to the right while the rules simple they cabple of preducing complex behivieur



model future secnion of market

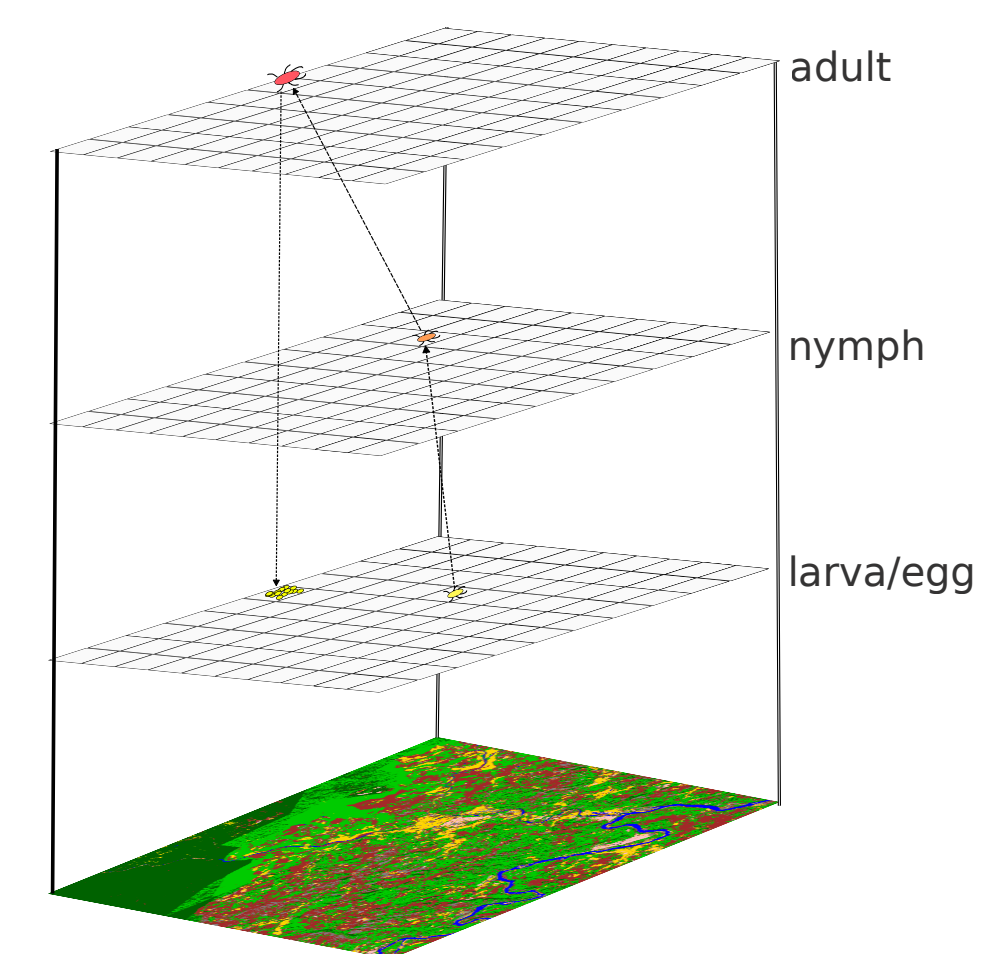
intgration

in north east madagascar market integration driving transition from shifted cultivation to agroforestry of vanilla, using CA we projected maps of future land use under diffrent level of market inegration and assumption about vanilla cultivation two example are:

- vanilla plot spread everywhere
- vanilla are mainly around the village

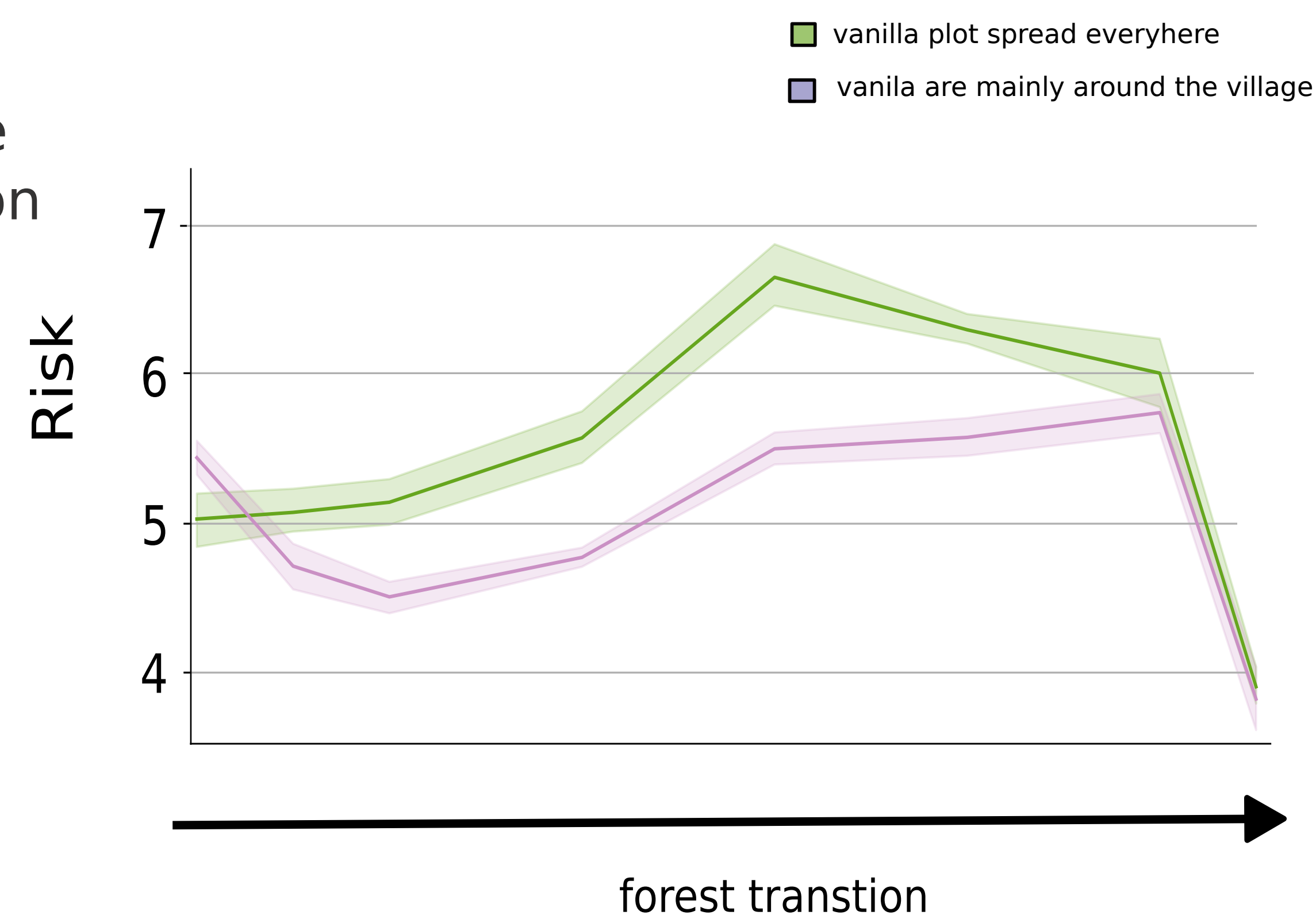
agent base model for ticks

after preducing the future land use map weuse agent base model to predict tick expousre in each one of those secnion, for this we create a simulation of tick population wit three layers each represnt tick lifestage



results

using those models we define four diffrent possialbe secnion for the transition



refreces

aknowlegment

